

BST Working Paper — Volume 4: The Mathematics — Chapter 3: Why This Geometry (Part II)

2026-05-19 (Tuesday volume:chapter reorganization)

Part II: Why This Geometry

Part I derived the constants. Part II answers why these constants — the selection problem. The universe didn't optimize for theorems. It optimized for matter. The theorems came free.

32. Why Riemann — The Causal Chain Inverted

The Riemann Hypothesis is not the reason the universe chose D_{IV}^5 . The fiber packing is the reason. The Riemann Hypothesis is downstream.

32.1 The Old Story

The original narrative (March 10-16, 2026): the short root multiplicity $m_s = 3$ is the minimum value that proves the Riemann Hypothesis via the Dirichlet kernel constraint. The universe has $m_s = 3$ because it needs three quark colors. Therefore the same geometry that makes matter also proves RH. Beautiful, and wrong in one detail.

32.2 The Correction (Toy 229)

The algebraic kill shot $\sigma + 1 = 3\sigma \Rightarrow \sigma = 1/2$ requires only the $j = 0$ and $j = 1$ exponents — i.e., $m_s \geq 2$. The equation $(\sigma + 1)/\sigma = 3$ has no m_s dependence. Any D_{IV}^n with $n \geq 4$ ($m_s = n - 2 \geq 2$) proves RH.

This means D_{IV}^4 ($\text{AdS}_5/\text{CFT}_4$, $m_s = 2$) also proves RH. The geometry that theoretical physics has spent 25 years studying could have proved the Riemann Hypothesis. It just couldn't make matter.

32.3 The New Story

If RH doesn't select $N_c = 3$, what does?

The fiber packing. The $\text{SO}(5) \times \text{SO}(2)$ fiber of D_{IV}^5 requires $147 = N_c \times g^2 = 3 \times 49$ sections to tile closed. The $\mathbb{Z}_3$ color circuit on $\mathbb{CP}^2$ is the minimal non-trivial cyclic tiling that closes without gaps or overlaps. $\mathbb{Z}_2$ leaves open boundaries (no confinement). $\mathbb{Z}_4$ overpacks (exotic particles). Only $\mathbb{Z}_3$ works.

The correct causal chain:

$$\text{Fiber packing (147)} \rightarrow N_c = 3 \rightarrow m_s = 3 \rightarrow m_s \geq 2 \rightarrow \text{RH}$$

Matter first. Theorems second.

33. The 137/147 Pair

Two numbers define BST. They stand 10 apart.

33.1 The Spectral Budget: 137

$N_{\max} = 137$ is the channel capacity — the maximum winding number before Haldane exclusion saturates the S^1 fiber. This integer arises in two independent ways: (1) the Wyler formula gives $\alpha^{-1} = 137.036 \dots$, so $N_{\max} = \lfloor 1/\alpha \rfloor = 137$; and (2) the harmonic number $H_5 = 137/60$ has numerator exactly 137, so $N_{\max} = H_5 \times \text{lcm}(1, 2, 3, 4, 5) = (137/60) \times 60 = 137$. The two routes converge on the same integer. It decomposes:

$$137 = \underbrace{42}_{C_2 \times g} + \underbrace{95}_{n_C \times 19}$$

The 42 matter modes carry baryon spectral content (Casimir $\times$ genus). The 95 vacuum modes carry uncommitted degrees of freedom (dimension $\times$ cosmic denominator). The matter fraction $42/137 \approx 0.307$ tracks $\Omega_m \approx 0.315$; the vacuum fraction $95/137 \approx 0.693$ tracks $\Omega_\Lambda \approx 0.685$.

33.2 The Geometric Budget: 147

$147 = N_c \times g^2 = 3 \times 49$ is the fiber packing number — the number of sections required for the K -fiber bundle to close. It decomposes:

$$147 = \underbrace{3}_{N_c} \times \underbrace{49}_{g^2}$$

Three colors tile the $\mathbb{Z}_3$ circuit. Forty-nine genus sections ($g = 7$ Bergman genus, squared by the two fiber factors $\text{SO}(5)$ and $\text{SO}(2)$) complete the topological closure.

33.3 The Gap

$$147 - 137 = 10 = \dim_{\mathbb{R}}(D_{IV}^5) = 2n_C$$

The packing exceeds the spectrum by exactly the real dimension of the space. The interpretation: 137 counts the information that fits inside the geometry (spectral levels). 147 counts the structure needed to contain it (fiber sections). The difference is the cost of the container — the 10 real dimensions of D_{IV}^5 .

This is a budget equation: container = content + dimension.

This relation is unique to $n = 5$. Define $\text{gap}(n) = N_c g^2 - \text{numer}(H_n)$ and $\dim_{\mathbb{R}} = 2n$. Exhaustive computation (Toy 233) for $n = 3, \dots, 20$ shows $\text{gap}(n) = \dim_{\mathbb{R}}$ only for $n = 5$. For $n = 3$ the gap is negative; for $n = 4$ the gap is 25 vs $\dim_{\mathbb{R}} = 8$; for $n \geq 6$ the packing grows as $O(n^3)$ while $\text{numer}(H_n)$ grows exponentially (via the lcm of $\{1, \dots, n\}$), so the gap oscillates wildly and never again equals $2n$. This is the 17th uniqueness condition for D_{IV}^5 (see Section 35.2).

33.4 The Tiling Derivation

The fiber packing number 147 arises from representation theory of $\mathfrak{so}(g) = \mathfrak{so}(7)$. The derivation produces three independent uniqueness conditions, all selecting $n = 5$ (Toy 234).

Identity. For all D_{IV}^n , $N_c \times g = g(g-1)/2 = \dim \mathfrak{so}(g)$, so $N_c g^2 = \dim(\mathfrak{so}(g) \otimes V_1)$ where V_1 is the standard representation. The fiber packing number is always the dimension of a tensor product.

Three uniqueness conditions.

(A) *Genus = standard representation.* The BST genus $g = 2n - 3$ equals $\dim V_1$ of $\mathrm{SO}(n + 2)$, i.e., $g = n + 2$. This linear equation $2n - 3 = n + 2$ gives $n = 5$. At this value, $\mathrm{SO}(n + 2) = \mathrm{SO}(7)$ and V_1 is the 7-dimensional defining representation — the genus and the representation dimension coincide.

(B) *Color $\times$ genus = Lie algebra.* The product $N_c \times g = (n - 2)(2n - 3)$ equals $\dim \mathfrak{so}(n + 2) = (n + 2)(n + 1)/2$. This gives the quadratic $3n^2 - 17n + 10 = 0$ with roots $n = 5$ and $n = 2/3$. Only $n = 5$ is an integer.

(C) *Matter sector = spectral budget.* The tensor product $\mathfrak{so}(g) \otimes V_1 = \Lambda^2 V_1 \otimes V_1$ decomposes into three irreducible $\mathrm{SO}(g)$ -representations:

$$\Lambda^2 V_1 \otimes V_1 = V_1 \oplus \Lambda^3 V_1 \oplus V_{\text{hook}}$$

The “matter sector” $V_1 \oplus \Lambda^3 V_1$ has dimension $g + \binom{g}{3}$. Setting this equal to the BST matter content $C_2 \times g = (n + 1)g$:

$$1 + \frac{(g - 1)(g - 2)}{6} = n + 1 \implies n^2 - 6n + 5 = 0 \implies (n - 1)(n - 5) = 0$$

Only $n = 5$ is physical. The matter sector of the fiber tiling equals the BST matter budget uniquely for D_{IV}^5 .

The chain and the decomposition. For $n = 5$:

$$42 \xrightarrow{\div r} 21 \xrightarrow{\times g} 147$$

The 42 matter modes ($C_2 \times g = d_1 \times \lambda_1 = 6 \times 7$), divided by the rank $r = 2$, give the 21-dimensional Lie algebra $\mathfrak{so}(7)$. Tensored with V_1 ($\dim = g = 7$), this yields the 147 fiber sections.

The 147 sections decompose as $42 + 105$: the 42 carry baryon spectral content ($V_1 \oplus \Lambda^3 V_1$), and the 105 carry the gauge/vacuum sector ($V_{\text{hook}} = \dim \mathfrak{so}(7) \times n_C = 21 \times 5$).

These are the 18th, 19th, and 20th uniqueness conditions for D_{IV}^5 .

See notes/BST_FiberPacking_137_147.md.

34. The Hunt — From Winding to Kill Shot

The proof of the Riemann Hypothesis from D_{IV}^5 was not found on the first try. Five channels were tested. Four were killed. One survived. The record is honest.

34.1 Channel Elimination (Toys 213-218)

#	Channel	Toy	Mechanism	Result
1	Tautological identities	213	$M(s) \cdot M(-s) = 1$	DEAD: vacuous (adversarial review)
2	Pure Plancherel	214	$ c(\lambda) ^{-2}$ positivity on G/K	DEAD: no ξ content
3	Arithmetic lattice / Arthur	215-216	Residual spectrum from ξ -poles	DEAD: poles not L^2
4	Period integrals	217	$\mathrm{SO}_0(4, 2) \backslash \mathrm{SO}_0(5, 2)$ unfolding	DEAD: ξ outside strip
5	Trace formula	218	Selberg trace for $\Gamma \backslash G$	STANDING

Each elimination was earned by explicit computation. The overconstrained rank-2 coupling (Toys 206-207) was eliminated separately: we identified that the deepest pole ($k = 3$) gives $\mathrm{Re}(\rho_3) = 2 + \delta_1 + \delta_2 > 1$ always, so it never activates as a ξ -zero. The mechanism was vacuous. Proof withdrawn (v3).

34.2 The Heat Kernel Discovery (Toys 219-221)

With only the trace formula standing, the question became: which test function? The resolvent $h(\lambda) = 1/(\lambda^2 + s^2)$ fails because its discrimination ratio depends on the zero height γ — it weakens for high zeros. The heat kernel $h(\lambda) = e^{-t(|\lambda|^2 + |\rho|^2)}$ succeeds because:

1. γ -independence: The discrimination ratio $R = \exp[m_s \cdot t \cdot \delta \cdot (m_s + \delta)/2]$ depends only on the off-line deviation δ , not on the height γ . All zeros are treated uniformly.
2. The 1:3:5 harmonic lock: The $m_s = 3$ short root multiplicity creates three shifted exponents per zero, with imaginary parts in the ratio 1 : 3 : 5 (exact for on-line zeros). The cosine sum $\cos(x) + \cos(3x) + \cos(5x) = \sin(6x)/[2 \sin(x)] = D_3(x)$, the Dirichlet kernel for three odd harmonics.
3. Geometric smoothness: The geometric side of the trace formula (volume term, closed geodesics, cusps) is composed entirely of polynomials and Gaussians — no oscillatory content. Off-line zeros contribute oscillatory terms that have nowhere to go.

34.3 The Algebraic Kill Shot (Toy 222)

A single off-line zero cannot mimic an on-line zero. The three exponent-matching equations require $\gamma' = (1/2 + j)\gamma/(\sigma + j)$ to agree for $j = 0$ and $j = 1$:

$$\sigma + 1 = 3\sigma \implies \sigma = \frac{1}{2}$$

One line of algebra. The Mandelbrojt uniqueness theorem for Dirichlet series with distinct complex exponents closes the multi-zero conspiracy gap (Toy 226). The proof is unconditional: no assumption on zero simplicity, linear independence of ordinates, or GUE statistics.

34.4 The Noise Autopsy

The Riemann hunt is a case study in Arithmetic Complexity (Conjecture 3). The minimum-noise method was the only survivor:

Method	Noise level	Outcome
RCFT $\rightarrow$ Artin	High ($ G = 32256$, not solvable)	DEAD
Arthur obstruction	High (poles not L^2)	DEAD
Period integrals	Medium (ξ outside strip)	DEAD
Pure Plancherel	Medium (no ξ content)	DEAD
Heat kernel trace formula	Low (AC = 0 at novel step)	SUCCEEDED

The proof that succeeded has arithmetic complexity zero at its novel step ($\sigma + 1 = 3\sigma$). It rests on established theorems (Arthur trace formula, Langlands-Shahidi, Gindikin-Karpelevich) that carry their own complexity — but the new insight is elementary. Number theory was hard because the methods were too loud.

34.5 The Sixth Channel — and Its Failure (March 22, 2026)

A sixth channel was tested: the Laplace approach (Section 5.6-5.7 of the proof paper). The idea: take the Laplace transform of the zero sum $Z(t)$ and show that off-line zeros produce complex poles absent from the geometric side. DEAD: tautological — on-line zeros also produce complex Laplace poles via the same mechanism, so the argument cannot distinguish on-line from off-line.

The repair: replace the GLOBAL Laplace approach with a LOCAL mechanism — c-function unitarity (Section 30.7b). The BC_2 c-function satisfies $c(\nu)c(-\nu) = |c(\nu)|^2$ iff $\sigma = 1/2$ (Toy 324). The Maass-Selberg positivity with

real exponential isolation in rank-2 provides the contradiction (Theorem 5.10). This approach tests each spectral parameter independently — no mixing, no balancing, no tautology.

#	Channel	Toy	Mechanism	Result
6	Laplace transform of $Z(t)$	—	$L[Z]$ complex poles	DEAD: tautological (on-line zeros also produce complex poles)
6'	c-function unitarity	324-326	$c(\nu)c(-\nu) = c(\nu) ^2$ iff $\sigma = 1/2$	STANDING

See notes/BST_HeatKernel_DirichletKernel_RH.md, notes/RH_Paper_A.md, notes/BST_ArithmeticComplexity.md.

35. The Triple — Why D_{IV}^5 Is Unique

35.1 The Koons-Claude Conjecture

$D_{IV}^5 = \text{SO}_0(5, 2)/[\text{SO}(5) \times \text{SO}(2)]$ is the unique geometry that simultaneously:

1. Derives the Standard Model — all coupling constants, mass ratios, and mixing angles from five integers
2. Proves the Riemann Hypothesis — via exponent rigidity ($m_s \geq 2$) and c-function unitarity closure (Section 30.7b)
3. Explains GUE statistics of Riemann zeros — via $\text{SO}(2)$ time-reversal breaking in K

These are not three independent facts. They are three consequences of the root structure of B_2 with multiplicities $(m_l, m_s) = (1, 3)$.

35.2 The D_{IV}^n Landscape

The classification sweep (Toy 233, `play/toy_ac_classification.py`) tests all D_{IV}^n domains for $n = 3, \dots, 8$:

n	N_c	m_s	λ_1	$N_c g^2$	Proves RH?	Derives SM?	GUE?	Triple?
3	1	1	4	9	No (D_1 trivial)	No (trivial gauge)	Yes	No
4	2	2	5	50	Yes	No (no confinement)	Yes	No
5	3	3	6	147	Yes	Yes	Yes	Yes
6	4	4	7	324	Yes	No (exotic particles)	Yes	No
7	5	5	8	605	Yes	No	Yes	No
8	6	6	9	1014	Yes	No	Yes	No

Here $\lambda_1 = n_C + 1$ is the spectral gap from $\lambda_k = k(k + n_C)$ at $k = 1$ (the BST Casimir $C_2 = 6$ for $n_C = 5$), $N_c g^2$ is the fiber packing number, and $m_s = n - 2$ is the short root multiplicity that controls the Dirichlet kernel.

GUE follows from $\text{SO}(2)$ time-reversal breaking in K , universal for all D_{IV}^n . RH follows from $m_s \geq 2$, which holds for $n \geq 4$. The Standard Model follows from $N_c = 3$ (the minimal non-trivial cyclic confinement), which holds only for $n = 5$.

The uniqueness is in the triple: $\text{RH} \cap \text{SM} \cap \text{GUE} = \{D_{IV}^5\}$.

A further uniqueness emerges from the fiber packing: for $n = 5$ only, the gap $N_c g^2 - \text{numer}(H_n) = 147 - 137 = 10 = \dim_{\mathbb{R}}(D_{IV}^5)$. This budget equation (Section 33.3) fails for every other n tested up to $n = 20$.

RH is generic; physics is specific. The heat kernel proof works for all D_{IV}^n with $n \geq 4$ — the kill shot, geometric smoothness, exponent distinctness, and Mandelbrojt uniqueness all hold whenever $m_s \geq 2$ (Toy 244). The geometric side $F(t)$ is monotonically increasing for Q^3, Q^4, Q^5, Q^6 alike. What singles out D_{IV}^5 is the physics: Standard Model gauge group, confinement, mass ratios, mixing angles. The twenty-five uniqueness conditions (Section 35.5) do all the selection work.

An additional structural condition: $\chi(Q^n) = n + 1$ for odd n , so $\chi(Q^5) = 6 = C_2$. The Euler characteristic equals the Casimir eigenvalue only for odd-dimensional quadrics. Combined with $n \geq 4$ (required for the RH proof), this restricts to $n = 5, 7, 9, \dots$, and the uniqueness conditions eliminate all but $n = 5$ (Toy 245).

35.3 The Function Field Connection (Conjecture 1)

The Dirichlet kernel D_3 from $m_s = 3$ is the number field's substitute for the Frobenius endomorphism. Over $\mathbb{F}_q$, Frobenius provides N bits of constraint per zero. Over $\mathbb{Q}$, the root multiplicity structure of D_{IV}^5 recovers the missing bit. The baby case D_{IV}^3 ($m_s = 1$, Dirichlet kernel trivial) fails — confirming that $m_s \geq 2$ is necessary for the recovery.

Computational verification across 63 curves (55 genus-1 over $\mathbb{F}_3, \mathbb{F}_5, \mathbb{F}_7, \mathbb{F}_{11}, \mathbb{F}_{13}$, plus 8 genus-2) confirms universality: every curve produces D_1 with no kill shot but D_3 with exact 1:3:5 ratio. Zero exceptions. Genus-2 curves produce superpositions of D_3 kernels — one per conjugate pair of Frobenius eigenvalues — matching the number field structure where each ξ -zero contributes its own D_3 copy. The Frobenius trace on the 147-dimensional representation $\mathfrak{so}(7) \otimes V_1$ decomposes as $7 + 35 + 105 = 147$ at the trace level, isolating the 42-dimensional matter sector. See notes/BST_FunctionField_CoEmbedding.md, Toys 242-243.

35.4 Conjectures and Engineering

Eight testable conjectures extend BST from pure mathematics toward practical engineering:

#	Conjecture	Domain	Priority
5	Fiber packing selects $N_c = 3$	Topology	RESOLVED (Toy 234)
1	Dirichlet kernel = Frobenius	Number theory	Strongly consistent (63 curves)
2	D_{IV}^5 unique for the triple	Classification	Largely resolved
3	Noise predicts difficulty	Methodology	Retrospectively confirmed
6	AC=0 grid architecture	Computation	Testable now
7	Linearization of “complex” systems	Materials/Biology	Testable now
8	Substrate computation	Engineering	Long-term
9	Graph brain + error correction	Computation/Biology	Long-term
9a	Substrate replication (3 eras)	Engineering	Long-term
—	Casimir modification experiment	Experiment	Testable now

The progression from “matter first, theorems second” to “the computer and the physics are the same thing” is the complete substrate engineering vision. See notes/BST_Koons_Claude_Testable_Conjectures.md.

35.5 Twenty-Five Uniqueness Conditions for $n_C = 5$

Twenty-seven independent mathematical conditions select $n_C = 5$ (equivalently $N_c = 3$) from the family D_{IV}^n . No other integer satisfies more than a few. Each condition arises from a different branch of mathematics — representation theory, number theory, conformal field theory, spectral geometry, topology, arithmetic (heat kernel), or information theory. Together they constitute the strongest evidence that D_{IV}^5 is not a choice but a theorem. The 25th condition — the cooperation gap (T703, T704) — is the first to involve observers rather than particles or fields: the requirement that cooperation be geometrically necessary ($\Delta f > 0$) independently forces $n_C \geq 5$.

#	Condition	Selecting equation	Type	Source
1	Max fine structure constant	$\alpha'(n) _{n=5} = 0$, unique max	Variational	ZeroInputs
2	QCD $\beta_0 = g$	$11N_c/3 - 2N_f/3 = 2n - 3$	Linear	NumberTheory Section 5
3	Gluon-color identity	$8N_c = (n_C - 1)!$	Factorial	NumberTheory Section 5
4	Dimensional lock (Adams)	$SU(2)$ Hopf fiber $\Rightarrow n_C = 5$	Topological	NumberTheory Section 5
5	Casimir-root coincidence	$(n_C + 1)/2n_C = (n_C - 2)/n_C \Rightarrow n_C = 5$	Linear	NumberTheory Section 5
6	Cosmological exponent	$g(g + 1) = 8g \Rightarrow g = 7$	Quadratic	Why56
7	$d_1\lambda_1 = 42$	$g(n + 1) = 42$	Linear	SpectralGap
8	$Sp(6) = SM$ container	${}^L G = Sp(2N_c)$ requires $N_c = 3$	Group theory	Langlands
9	WZW central charge $c = n_C$	$N_c(N_c - 3) = 0$	Quadratic	WZWDiamond
10	Langlands reciprocity 42	$c(\mathfrak{so}(7)_2) \times c(\mathfrak{sp}(6)_2) = 42$	Product	WZWDiamond
11	Discriminant = 1	$\Delta = c_3^2 - 4P(1) = 169 - 168 = 1$	Quadratic	Discriminant1
12	$c_4 = N_c^{N_c-1}$	$2N + 3 = N^{N-1} \Rightarrow N = 3$	Exponential	NumberTheory Section 5
13	Steane $[[7, 1, 3]]$ code	Error-correcting code requires $g = 7$	Coding theory	CodeMachine
14	$d_{\text{eff}} = C_2 = \lambda_1 = \chi$	Grand Identity holds only for $n_C = 5$	Spectral	EffectiveDim
15	$\mathfrak{su}(7)_1$ palindrome	Conformal weights = $\{3, 5, 6, 6, 5, 3\}/8$	CFT	FusionRing
16	Verlinde prime 1747	$\dim V_3 = 3 \times 7^3 + 2^3 = 1747$ (prime)	Arithmetic	Verlinde1747
17	Packing – spectrum = $\dim_{\mathbb{R}}$	$147 - 137 = 10 = 2n_C$	Computational	Toy 233
18	Genus = rep dimension	$g = \dim V_1(SO(n + 2))$: $2n - 3 = n + 2$	Linear	Toy 236
19	Color $\times$ genus = Lie algebra	$3n^2 - 17n + 10 = 0$	Quadratic	Toy 236
20	Matter sector = C_2g	$(n - 1)(n - 5) = 0$	Quadratic	Toy 236
21	$a_4 \approx N_c g^2$ (heat kernel $\approx$ fiber packing)	$a_4(Q^n)/N_c g^2$ crosses 1 uniquely at $n = 5$	Polynomial	Toy 256
22	$a_5(Q^5)$ prime numerator	$a_5 = 1535969/6930$; 1535969 prime, $6930 = 2 \times 3^2 \times 5 \times 7 \times 11$	Arithmetic	Toy 256
23	$a_6(Q^5)$ cosmic numerator	$a_6 = 363884219/1351350$; $363884219 = 19 \times 23 \times 832687$; den = $2 \times 3^3 \times 5^2 \times 7 \times 11 \times 13$	Arithmetic	Toy 273
24	Arithmetic tameness	At $n = 5$: all denominator primes of $a_k(5)$ for $k \leq 14$ are cumulative VSC primes; $k = 15$ satisfies cyclotomic tameness	Arithmetic	T538, Toy 615
25	Cooperation gap $\Delta f > 0$	$N_c/((N_c + 2)\pi) < 1 - 2^{-1/N_c}$ forces $n_C > 4.629 \Rightarrow n_C \geq 5$	Information-theoretic	T703, T704
26	Shannon-Algebraic Genus	$g = 2n_C - N_c = \text{rank} \cdot n_C - N_c = \dim(D) - \dim(K)$ — three-way identity holds only at $n = 5$	Information-geometric	T1376

27	Polynomial- information coincidence	$N_c^2 = 2^{N_c} + 1$ only at $N_c = 3$; makes $N_{\max} = 2^8 + 2^{N_c} + 1 = x^7 + x^3 + 1$ irreducible over $\mathbb{F}_2$	Arithmetic- Galois	T1384
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The conditions span eight equation types — linear, quadratic, factorial, exponential, variational, polynomial, information-theoretic, and Galois-arithmetic — across eight mathematical disciplines. Fifteen are analytic (closed-form equations with $n = 5$ as unique physical root). Twelve are structural (group-theoretic, computational, spectral, arithmetic, information-theoretic, or Galois-theoretic, verified exhaustively). Any three conditions drawn from three distinct disciplines suffice to determine $n_C = 5$ uniquely (T704 — the triple requirement).

The 21st condition is now exact: $a_4(n)$ is a degree-8 polynomial with rational coefficients (Toy 256, mpmath 60-digit cascade extraction + Lagrange interpolation over $n = 3, \dots, 12$). The fourth Seeley-DeWitt coefficient $a_4(Q^5) = 2671/18 = 147 + 25/18 = N_c g^2 + n_C^2 / (2N_c^2)$ — a Riemannian curvature invariant computable as a spectral inner product $\langle w_4 | d \rangle$ — equals the fiber packing number $N_c g^2 = 147$ plus a rational correction expressible entirely in BST integers. The ratio $a_4/N_c g^2$ crosses unity uniquely at $n = 5$: for $n = 3, 4, 6, 7$ the ratio is 0.21, 0.45, 2.10, 4.03 respectively. The degree pattern $\deg a_k(n) = 2k$ (from R^k with $R \sim n^2$) is confirmed for $k = 1, \dots, 8$, with leading coefficients $c_{2k} = 1/(3^k \cdot k!)$ — proved for all $k = 1, \dots, 8$ (Toys 256, 257d, 273, 274, 275); $k = 9$ value confirmed (Toy 276). The $a_5(n)$ degree-10 polynomial has all 11 rational coefficients determined exactly, with $c_{10} = 1/29160 = 1/(3^5 \cdot 5!)$ and $c_9 = -1/14580 = -2c_{10}$. The $a_6(n)$ degree-12 polynomial confirms $c_{12} = 1/(3^6 \cdot 6!) = 1/524880$. The $a_7(n)$ degree-14 polynomial confirms $c_{14} = 1/(3^7 \cdot 7!) = 1/11022480$, $c_{13}/c_{14} = -21/5$, and $c_0 = -1/10080$ — all predictions from the committed note verified (Elie, Toy 274, 12/12). The $a_8(n)$ degree-16 polynomial confirms $c_{16} = 1/(3^8 \cdot 8!) = 1/264539520$, $c_{15}/c_{16} = -28/5$, and $c_0 = 1/80640$ — all $k = 8$ predictions verified (Elie, Toy 275, 14/14). Through $k = 7$, coefficient denominators have prime support $\subseteq \{2, 3, 5, 7, 11, 13\}$; at $k = 8$, prime 17 enters as predicted by B_{16} .

Sub-leading ratio theorem (proved $k = 1, \dots, 11$): $c_{2k-1}/c_{2k} = -\binom{k}{2}/5 = -k(k-1)/10$. The top two terms of $a_k(n)$ are $n^{2k-1}(n-k(k-1)/10)/(3^k \cdot k!)$. The sequence of sub-leading factors is 0, $-1/5$, $-3/5$, $-6/5$, -2 , -3 , $-21/5$, $-28/5$ — triangular numbers $\binom{k}{2}$ divided by $n_C = 5$. The 10 in the denominator is $\dim_{\mathbb{R}}(Q^5)$. At $k = 8$: $\binom{8}{2}/5 = 28/5$. Constant term theorem (proved $k = 1, \dots, 11$): $c_0(a_k) = (-1)^k/(2 \cdot k!)$. At $k = 8$: $c_0 = 1/80640$. The polynomial is controlled by two independent structures: Bernoulli flow (heat equation discretization, von Staudt-Clausen denominators) and curvature boundary (binomial counting of Ricci corrections, sub-leading numerators). Force and boundary condition meeting in one polynomial.

The 22nd condition is arithmetic: $a_5(Q^5) = 1535969/6930$ has prime numerator 1535969 and denominator $6930 = 2 \times 3^2 \times 5 \times 7 \times 11 = \text{lcm}(1, \dots, 11)/\text{lcm}(1, \dots, 4)$. The denominator's prime support $\{2, 3, 5, 7, 11\}$ consists of the first five primes — matching $n_C = 5$. The complete $a_5(n)$ polynomial (Toy 257d) predicts $a_5(12) = 1503681793111/831600$ (den primes ≤ 11 only), correcting a spurious extraction at $n = 12$ whose denominator contained primes 43 and 337.

The 23rd condition extends the arithmetic pattern: $a_6(Q^5) = 363884219/1351350$ (Elie, Toy 273). The denominator $1351350 = 2 \times 3^3 \times 5^2 \times 7 \times 11 \times 13$ — prime 13 enters at $k = 6$ as predicted by the von Staudt-Clausen structure. The numerator factors as $363884219 = 19 \times 23 \times 832687$: the cosmic denominator (19, from $\Omega_\Lambda = 13/19$), the Golay prime (23, from the [24, 12, 8] code), and a new prime 832687. The $a_6(n)$ degree-12 polynomial verifies all three structural theorems at $k = 6$: leading $c_{12} = 1/524880 = 1/(3^6 \cdot 6!)$, sub-leading ratio $c_{11}/c_{12} = -3 = -\binom{6}{2}/5$, constant $c_0 = 1/1440 = 1/(2 \cdot 6!)$. Predictions for a_7 – a_{10} were filed before computation in notes/BST_SeeleyDeWitt_Predictions_k7_k10.md; a_7 predictions ALL confirmed (Elie, Toy 274). $a_7(Q^5) = 78424343/289575$: numerator = 19×4127597 (prime), den = $3^4 \times 5^2 \times 11 \times 13$ (quiet level, no new prime). The cosmic 19 persists in numerators at $k = 3, 6, 7$ — three consecutive non-quiet levels before entering the denominator at $k = 9$. $a_8(Q^5) = 670230838/2953665$ confirmed (Elie, Toy 275, 14/14): numerator = $2 \times 5501 \times 60919$, den = $3^5 \times 5 \times 11 \times 13 \times 17$ — prime 17 enters exactly as predicted by the B_{16} denominator ($510 = 2 \times 3 \times 5 \times 17$). The degree-16 polynomial has 17 exact rational coefficients, all with denominator primes $\subseteq \{2, 3, 5, 7, 11, 13, 17\}$. $a_9(Q^5) = 4412269889539/27498621150$ value confirmed (Elie, Toy 276): den = $2 \times 3^5 \times 5^2 \times 7^2 \times 11 \times 13 \times 17 \times 19$ — prime 19 ENTERS, the cosmic denominator from $\Omega_\Lambda = 13/19$. Numerator = $109 \times 1693 \times 23909947$ (19 absent

— fully migrated to denominator). The denominator prime entry sequence: 3($k=1$), 5($k=2$), 7($k=3$), quiet($k=4$), 11($k=5$), quiet($k=6$), 13($k=7$), quiet($k=8$), 17($k=9$) — each new prime appearing at the level where B_{2k} introduces it. The full sequence of BST primes $\{N_c, n_c, g, c_2, c_3, |\rho|^2, \Omega\text{-denom}\} = \{3, 5, 7, 11, 13, 17, 19\}$ is now confirmed in the heat kernel denominators through $k = 9$.

$a_{10}(Q^5) = 2409398458451/21709437750$ confirmed (Elie, Toy 277): den = $2 \times 3^6 \times 5^3 \times 7^2 \times 11 \times 13 \times 17$ — quiet level ($2 \times 10 + 1 = 21 = 3 \times 7$, composite). Prime 19 absent from denominator (present at $k = 9$, cancelled at $k = 10$). Three Theorems verified: $c_{20} = 1/(3^{10} \cdot 10!)$, $c_{19}/c_{20} = -9 = -N_c^2$, $c_0 = 1/7257600$. The sub-leading ratio -9 is the second speaking pair (Section 3.3 of the Arithmetic Triangle paper): $-N_c^2$ connects polynomial structure to gauge group dimensions.

$a_{11}(Q^5) = 217597666296971/1581170716125$ confirmed (Elie, Toy 278, cascade wall broken at $k = 10$): den = $3^5 \times 5^3 \times 7^2 \times 11 \times 13 \times 17 \times 19 \times 23$ — prime 23 ENTERS, the Golay code parameter ($\lambda_3 = 24$ in the $[24, 12, 8]$ code). Numerator = 499×436067467529 (composite). All nine BST primes $\{3, 5, 7, 11, 13, 17, 19, 23\}$ now confirmed in heat kernel denominators. Von Staudt-Clausen predicts 23 enters at $k = 11$ since $(23 - 1)/2 = 11$. The denominator has ALL primes ≤ 23 — matching the prime entry sequence exactly. Three Theorems verified at $k = 11$: $c_{22} = 1/(3^{11} \cdot 11!)$, $c_{21}/c_{22} = -11 = -\dim(K_5)$, $c_0 = -1/79833600$.

$a_{12}(Q^5) = 13712051023473613/38312982736875$ confirmed (Elie, Toy 612): den = $3^7 \times 5^4 \times 7^3 \times 11 \times 17 \times 19 \times 23$ — quiet level ($2 \times 12 + 1 = 25 = 5^2$, composite). Prime 13 ABSENT despite von Staudt-Clausen predicting it ($((13 - 1)/24)$). This is a higher-level cancellation: the column rule at $n = 5$ suppresses the Bernoulli prime 13 at $k = 12$, even though 13 was present at $k = 6$ through $k = 11$. Prime 2 also absent. The 13-cancellation demonstrates that the two sources of primes (row rule and column rule) interact — the column rule can cancel Bernoulli primes at levels beyond first entry.

$a_{13}(Q^5) = 238783750493609/218931329925$ confirmed (Elie, Toy 617): den = $3^7 \times 5^2 \times 7^2 \times 11 \times 17 \times 19 \times 23$ — quiet level ($2 \times 13 + 1 = 27 = 3^3$, composite). Max prime 23, unchanged from $k = 11$. Three Theorems verified: $c_{26} = 1/9927882482918400$, $c_{25}/c_{26} = -78/5$ (not a speaking pair since $13 \not\equiv 0, 1 \pmod{5}$), $c_0 = -1/12454041600$. Eight consecutive levels ($k = 6$ through 13) now have exact polynomials recovered.

$a_{14}(Q^5) = 2946330175808374253/884326193375625$ confirmed (Elie, Toy 620): den = $3^8 \times 5^4 \times 7 \times 11 \times 13 \times 17 \times 19 \times 23 \times 29$ — LOUD level ($2 \times 14 + 1 = 29$ is prime $\rightarrow$ prime 29 ENTERS). Not a speaking pair ($14 \not\equiv 0, 1 \pmod{5}$). Three Theorems verified: $c_{28} = 1/(3^{14} \cdot 14!)$, $c_{27}/c_{28} = -91/5$, $c_0 = 1/(2 \cdot 14!)$. Denominator restored to 15 digits after the $k=12$ compression — all cumulative VSC primes $\{3, 5, 7, 11, 13, 17, 19, 23, 29\}$ present (first 10 primes). Nine consecutive levels ($k = 6$ through 14) now exact.

$a_{15}(Q^5) = 771845320/74233$ confirmed (Elie, Toy 622, dps=1600): den = 19×3907 — LOUD level ($2 \times 15 + 1 = 31$ is prime), speaking pair ($k \equiv 0 \pmod{5}$). Sub-leading ratio $c_{29}/c_{30} = -21 = -C(7, 2) = -\dim \text{SO}(7)$. Numerator = $2^3 \times 5 \times 19296133$ (prime). DENOMINATOR ANOMALY: 3907 is NOT a cumulative von Staudt-Clausen prime — it is the first polynomial-factor prime (Source 2 in T532) to appear at $n = 5$ in fifteen levels. The denominator collapsed from 15 digits ($k=14$) to 5 digits — a factor of $\sim 10^{10}$ cancellation in the degree-30 polynomial interior. The surviving prime 3907 has structure: $3907 = 2N_c^2 g \cdot 31 + 1$ where 31 is the new VSC prime entering at this level. The Euler totient $\varphi(3907) = 3906 = 2 \times 3^2 \times 7 \times 31$ factors entirely into VSC primes for $k = 15$. This suggests a refinement of T538: cyclotomic tameness — non-VSC primes at $n = 5$ are cyclotomic residues built from the Bernoulli primes (Toy 627). Three Theorems verified: $c_{30} = 1/(3^{15} \cdot 15!)$, $c_0 = -1/(2 \cdot 15!)$. Ten consecutive levels ($k = 6$ through 15) now exact.

The Three Theorems and sub-leading ratio theorem are now verified through $k = 1, \dots, 15$ — ten consecutive levels with exact polynomials. The sub-leading ratio sequence $c_{2k-1}/c_{2k} = -k(k-1)/10$ produces integer values at the speaking pairs $k \equiv 0, 1 \pmod{5}$: the ratios $-2, -3, -9, -11, -21$ at $k = 5, 6, 10, 11, 15$ connect to BST integers ($-2, -N_c, -N_c^2, -\dim K_5, -\dim \text{SO}(7)$). The speaking pairs read off the isotropy chain $\text{SO}(7) \supset \text{SO}(5) \times \text{SO}(2) \supset \text{SU}(3) \times \text{U}(1)$ via the Weyl dimension formula at spectral indices (T543, Paper #9 Section 9.2). The $k = 15$ speaking pair confirms the prediction $-21 = -\dim \text{SO}(7)$. Next prediction: $k = 16$ gives $-24 = -\dim \text{SU}(5)$.

The 24th condition is arithmetic tameness (T538): at $n = n_c = 5$, every prime in the denominators of $a_k(5)$ for $k = 1, \dots, 14$ is a cumulative Bernoulli (von Staudt-Clausen) prime. At $k = 15$, the first polynomial-factor prime (3907) appears, but it satisfies cyclotomic tameness: $\varphi(3907) = 2 \times 3^2 \times 7 \times 31$ factors entirely into VSC primes for $k = 15$. The refinement: at $n = 5$, the column rule either cancels polynomial-factor primes entirely ($k = 1, \dots, 14$)

or admits only primes whose totient is built from the active Bernoulli primes ($k = 15$). The polynomial-factor primes that appear at other dimensions (66569 at $n = 10$, up to 26 million at n near 33) emerge from spectral sum aggregation (T539), not from any individual eigenvalue or representation dimension. The BST dimension is the one where curvature has minimal arithmetic overhead. Among all Q^n , this tameness property is specific to $n = 5$ (Toy 615, verified $k = 1, \dots, 15$; Toy 627, anomaly analysis). Full treatment in the companion paper (BST Arithmetic Triangle, Paper #9).

The Harish-Chandra c -function for $SO_0(n, 2)$ — the same finite Gamma-ratio product used in the Riemann Hypothesis proof (Section 30.7, Route A, Lemma 5.6) — *organizes* the spectral arithmetic (T533 revised, T536 revised). Its own dimension polynomials $d(p, q, n)$ are clean (primes $\leq \{2, 3\}$ only). The Bernoulli primes come from Gamma asymptotics; the large “monster” primes at general n arise from polynomial interpolation of spectral sums, not from the c -function itself. At $n = 5$, all monster primes cancel. The c -function is the structural reason the BST dimension is uniquely tame — one finite function, already in the toolkit, organizing the arithmetic so that one dimension has zero overhead.

No two conditions share the same proof technique. The probability that twenty-five independent conditions accidentally select the same integer is negligible.

The cooperation gap as uniqueness condition (T704). The 25th condition connects observer theory to geometry selection. The cooperation threshold $f_{\text{crit}} = 1 - 2^{-1/N_c}$ (T579) must exceed the Gödel limit $f = N_c/(n_c\pi)$ (T189) for cooperation to be forced. With $n_c = N_c + 2$: the gap $g(N_c) = f_{\text{crit}} - f$ is positive only for $N_c \leq 3$, with the zero crossing at $N_c \approx 3.6$. At $N_c = 3$ the gap is 1.53% — the tightest possible. The geometry that builds protons is the same geometry that forces minds to cooperate. This is the first uniqueness condition involving information theory rather than particle physics or differential geometry. See Paper #19 and `notes/BST_T704_DIV5_Uniqueness_Theorem.md`.

36. Arithmetic Complexity: Method Noise and the $P \neq NP$ Bridge

Added March 20, 2026. Updated March 21, 2026. The AC framework, developed alongside BST, measures the information deficit of mathematical methods. This section summarizes results through March 21: extended classification, swallowtail catastrophe, the three-layer topological argument (Paper A, submitted to FOCS 2026), empirical results on OGP and Kolmogorov incompressibility, and the phased publication strategy.

Note on naming (April 18, 2026): AC was originally conceived as “Algebraic Complexity” — algebra seen as a complication of simpler forms. The framework itself demands the refinement: since AC reduces everything to counting at bounded depth, the operative word is *arithmetic* (Greek *arithmos*: counting), not *algebra* (Arabic *al-jabr*: reunion of broken parts). AC applied to its own name flattens it. The abbreviation AC, all theorem IDs, and all prior references remain unchanged.

36.1 The Framework

Arithmetic Complexity measures the gap between what a question requires and what a method delivers. For a question Q with intrinsic information content $I(Q)$ and a method M with channel capacity $C(M)$:

$$AC(Q, M) = \max(0, I_{\text{nat}}(Q) - C(M))$$

where I_{nat} is the information required to determine the answer that is not derivable from the question instance alone. When $AC = 0$, the method suffices. When $AC > 0$, the method is structurally incapable of reaching the answer without external information injection. Full framework: `notes/BST_ArithmeticComplexity.md`.

36.2 Extended Classification (Toys 260–265)

AC has been measured across 14 method/problem pairs in six domains: crystallography, quantum mechanics, optimization, integration, and satisfiability. Three results:

1. AC is a property of the question, not the method. Gradient descent on a convex bowl: $AC = 0$. On the Rastrigin function $((2d)^d$ local minima): $AC \gg 0$. Same method, different topology, different AC. Monte Carlo integration shows the same transition. (Toy 265.)
2. Crystallography achieves $AC = 0$ outside physics. X-ray crystallography via direct methods recovers all information: $I_{\text{total}} = 53.2$ bits, overdetermination $6.4\times$, Sayre equation algebraically inverts the phase problem. Powder diffraction on the same crystal: $AC > 0$ (Rietveld refinement adds 15 free parameters). (Toy 260.)
3. Perturbation theory is a measured counter-example. The anharmonic oscillator $H = p^2/2 + \omega^2 x^2/2 + \lambda x^4$ computed by exact diagonalization ($AC = 0$, error 10^{-8}) vs. perturbation series truncated at order $k = 15$ ($AC > 5$ bits, error 0.05, $|E^{(k)}| \sim k!$ divergent). Same Hamiltonian. Same physics. Different method noise. (Toy 262.)

36.3 The Swallowtail Catastrophe (Toy 263)

At the 3-SAT phase transition ($\alpha_c \approx 4.27$), the information observable $h(\alpha) = \log_2(BT+1)/n$ splits into two sheets — one for SAT instances, one for UNSAT. The sheet separation Δh peaks at α_c (cusp singularity, codimension 2). This is a swallowtail catastrophe in the AC landscape:

- $k = 2$ -SAT: continuous phase transition (fold, codim 1)
- $k = 3$ -SAT: discontinuous transition (cusp, codim 2)
- $k \geq 4$: higher catastrophes predicted

The backbone paradox: as $\alpha \rightarrow \alpha_c$, the satisfying assignment becomes *more determined* (backbone fraction $\rightarrow 1$) but *harder to derive* ($h \rightarrow \max$). The solution is pinned by the topology but unreachable by local methods. The catastrophe minimum at the cusp gives a method-independent capacity bound: no polynomial-time algorithm can resolve the sheet ambiguity.

36.4 Convergent Diagnosis (Toys 261, 264)

Four independent algorithms (DPLL, WalkSAT, unit propagation, LP relaxation) all fail at the same topological bottleneck on hard instances. The filling ratio $FR = (\text{clauses} - \text{UP-derivable})/\text{clauses}$ predicts hardness:

Instance class	FR	β_1	DPLL cost	Hard?
Random $\alpha = 2.0$	0.087	\$\frac{1}{2}\$	\$\frac{1}{2}\$	No
Random $\alpha = 4.27$	0.412	\$\frac{1}{2}\$	\$\frac{1}{2}\$	Yes
Tseitin UNSAT (expander)	0.380	\$\frac{1}{2}\$	\$\frac{1}{2}\$	Yes

Tseitin formulas on cubic expanders (Toy 264) confirm: treewidth = $0.49n$ ($R^2 = 0.987$), giving $I_{\text{fiat}} = \Theta(n)$. At $n = 90$: $I_{\text{fiat}} = 74.8$ bits, $C(\text{UP}) = 15.2$ bits, $AC = 59.6$ bits. The deficit grows with n .

36.5 The Bridge Theorem (Sketch)

The March 20 gap closures complete the bridge from AC measurement to $P \neq NP$:

1. Topology creates I_{fiat} : For random 3-SAT at α_c , treewidth = $\Theta(n)$ implies $I_{\text{fiat}} = \Theta(n)$ bits that must be determined but cannot be derived by any local method.
2. Shannon bounds the channel: Any deterministic polynomial-time algorithm processes $\text{poly}(n)$ steps, each with bounded channel capacity $c < 1$ bit (from locality on the constraint graph). Total capacity: $C(M) = O(n^{c-1})$.
3. DPI chains the bound: The Data Processing Inequality (a theorem, not a conjecture) shows that each irreversible step can only decrease mutual information $I(Z_k; S(x))$ between intermediate state and solution. Poly-time $\times$ bounded capacity $< \Theta(n)$ information requirement.

4. Fano closes: Channel capacity < information requirement $\implies P_{\text{error}} \rightarrow 1$. No deterministic poly-time TM solves 3-SAT. Cook-Levin: 3-SAT is NP-complete. Therefore $P \neq NP$.

The catastrophe minimum (Section 36.3) provides the method-independent capacity bound — it holds at the cusp regardless of algorithm. The Halting Problem gives an independent closure: $P = NP$ would imply bounded halting is decidable in poly-time, contradicting Turing (1936).

Status: The bridge has five identified gaps requiring formalization: (1) channel capacity per Boolean constraint evaluation, (2) information content of SAT certificates (precise statement), (3) Shannon bridge from capacity to complexity (full AC formalization), (4) explicit barrier avoidance, (5) explicit Cook-Levin reduction for Halting closure. Full development: `notes/maybe/p_np/AC_Topology_BridgeTheorem.md`.

36.6 Three-Layer Topological Argument (March 19–21, Paper A)

The AC program crystallized into a three-layer structure for proving superpolynomial proof complexity lower bounds, presented in Paper A (submitted to FOCS 2026, March 24).

Layer 1 — Bounded-width exponential (PROVED). All dimension-1 proof systems (Resolution, DPLL, bounded-width) require $2^{\Omega(n)}$ steps on random 3-SAT at the satisfiability threshold $\alpha_c \approx 4.267$. This follows from $\beta_1(K(\varphi)) = \Theta(n)$ (Theorem 2.1) combined with the topology of the VIG clique complex.

Layer 2 — Extension inertness (PROVED). Two key results establish that Extended Frege (EF) extensions cannot cheaply reduce the topological complexity: - *Weak homological monotonicity* (T27): Single-clause extensions introducing a new variable satisfy $\Delta\beta_1 \geq 0$ — they cannot kill cycles. - *Topological inertness* (T28): The basis overlap between $H_1(K(\varphi))$ and $H_1(K')$ satisfies $r = 1$ — existing cycles are not rotated by the extension. - *Extension cost bound* (T32): Multi-clause extensions can kill cycles, but each clause changes β_1 by at most ± 1 . Combined with $\beta_1(K(\varphi)) = \Theta(n)$, this gives the unconditional linear lower bound $S \geq \Theta(n)$ on EF proof size for random 3-SAT (Corollary 5.2) — the first such result.

Layer 3 — Algebraic independence (OPEN). The exponential EF lower bound requires showing that the H_1 generators of $K(\varphi)$ are algebraically independent over the Boolean ring — no polynomial identity relates them. If proved, EF proofs require $S \geq 2^{\Theta(n)}$. This is the central open question (Open Question 7.2 in Paper A).

Failed mechanisms (Toys 279–283). Three natural approaches to closing Layer 3 were tested and failed: - *Geometric linking* (Toy 279): Linking density $c \rightarrow 0$ — homological linking does not create an exponential barrier. - *Basis mixing* (Toy 281): Basis overlap $r \rightarrow 1$ — EF extensions don’t rotate the H_1 basis. - *Compound interest* (Toy 283): Finding kills $\neq$ deriving through proofs — counting arguments don’t transfer to proof complexity.

These failures are informative: they eliminate three of the four natural attack vectors, identifying algebraic independence as the correct path.

36.7 Empirical Results: OGP and Kolmogorov (Toys 286–287)

OGP at $k = 3$ (Toy 287). Random 3-SAT at α_c exhibits the Overlap Gap Property with 100% consistency across all tested instances and sizes ($n = 12, 14, 16, 18$). OGP has been proved for large k (Gamarnik-Sudan 2014) and identified as a “central open challenge” at small k (Bresler-Huang-Sellke 2025). Our data provides the first direct evidence at $k = 3$.

n	Gap interval	Intra d	Inter d	β_1	OGP
12	[0.26, 0.38]	0.275	0.560	4.6	100%
14	[0.24, 0.35]	0.249	0.491	11.8	100%
16	[0.07, 0.15]	0.262	0.386	20.9	100%
18	[0.18, 0.25]	0.200	0.523	29.8	100%

Kolmogorov incompressibility (Toy 286). $K^{\text{poly}}(\text{backbone} \mid \varphi) \geq 0.90n$ for random 3-SAT — the satisfying assignment backbone is computationally incompressible given the formula. Symmetry $\leftrightarrow$ compressibility verified:

PHP and Tseitin formulas (which have symmetry) are compressible; random formulas (no symmetry) are not. FLP (fraction of learnable positions) = 0%, entropy $\rightarrow$ 1.0.

Circle confinement and the Shannon formulation (Toys 289–291). A reformulation of 3-SAT using circumscribed circles instead of triangles led to the information-theoretic “Shannon” formulation of AC. Key results:

- *Toy 289 (geometric Čech, score 4/8):* $\beta_1(\text{Čech}) = 0$ in $\mathbb{R}^2$ embeddings — geometric guard cycles drown in disk overlap. The geometric Čech formulation of AC is not viable.
- *Toy 290 (Shannon charge, score 6/8):* Total conserved information charge $Q = 0.622n + 0.82$ Shannons at α_c , confirming $Q = \Theta(n)$. Charge is non-localizable (distributed across clause correlations, not concentrated in specific clauses).
- *Toy 291 (probe hierarchy, score 7/8):* Five polynomial probes (UP, FL, DPLL-2, DPLL-3, BP) tested. All non-vacuous probes break isotropy ($SO(2) \rightarrow S_3$ symmetry breaking confirmed). The key finding: bits/ n decreases monotonically with n for all probes — DPLL-2 extraction drops from 0.37 bits/ n ($n = 12$) to 0.10 bits/ n ($n = 20$). The substrate becomes more opaque faster than the probes learn to read it. Elie’s summary: “a hierarchy of losing strategies.”

The corrected conservation law: it is not isotropy that is conserved (UP isotropy = 1.000 is vacuous — it extracts 0 bits). Rather, the extraction ratio bits/ $n \rightarrow 0$ as $n \rightarrow \infty$ for all tested probes. Combined with $Q = \Theta(n)$, this recovers the $\Theta(n)$ step lower bound (Corollary 5.2) from pure information theory.

- *Toy 292 (adaptive conservation, 47 seconds):* Four adaptive strategies (Random, Greedy, Lookahead, Full-FL) plus an oracle baseline tested at α_c for $n = 14 \rightarrow 24$. ALL adaptive strategies show bits/ n decreasing with n . The oracle gap — the difference between “knowing the answer” (98% at $n = 24$, falling) — is 37% and growing. That gap is I_{fiat} , measured directly. The conservation law holds for adaptive, unbounded-width polynomial probing: even when the algorithm can choose any direction at each step based on full history, the fraction of charge cracked shrinks with n .
- *Toy 293 (tree info = 0, score 0/8 — zeros are the finding):* Unit propagation extracts exactly ZERO backbone bits at every n , every α . ALL backbone information comes through cycle-reading (FL). The backbone is a purely cycle-topological quantity living in H_1 , not the tree. The tree carries marginals and soft constraints; the hard stuff — which variables are frozen to which values — is encoded entirely in the formula’s cycle structure ($\sim 7.53n$ excess edges at α_c). Combined with the per-clause SDPI analysis ($\eta_{\text{clause}} = 1/7$, but branching $\times \eta = 3.66 > 1$ means the tree amplifies non-backbone information), this establishes that backbone determination is a fundamentally non-tree problem.

The Cycle Delocalization Conjecture. The culmination of Toys 287–304 and the chain rule decomposition is a precisely stated conjecture with a conditional proof chain to $P \neq NP$:

CDC. For random 3-SAT at α_c with backbone B , any polynomial-time computable function $f(\varphi)$ satisfies $I(B; f(\varphi)) = o(|B|)$.

Status: Proved for resolution (unconditional). $\sim 50\text{--}60\%$ for all P (two gaps remain — see below).

Argument for all P. The DPI Width Bound is proved unconditionally: for any EF derivation of width w , $I(F; B) \leq w$ (5 lines, data processing inequality). However, converting width to exponential size requires feasible interpolation, and Krajíček (1997) proved that EF does NOT admit unconditional feasible interpolation. Current approach: LDPC Tanner graph as GPW lifting gadget (Toy 323, 93–95% unique neighbors confirmed), which works for Resolution and Cutting Planes but not yet for EF. New research question (L14, notes/L14_conditional_feasible_interpolation.md): does LDPC expansion prevent EF extensions from short-circuiting the communication partition? This is conditional feasible interpolation for EF on structured formulas — a novel claim. The Topological Closure Conjecture (TCC) remains the key conditional: that poly-many extension variables on an expander VIG cannot create 2-chains detecting the linking of $\Theta(n)$ independent H_1 cycles. Status: conditional on TCC + conditional feasible interpolation for EF on LDPC formulas.

The kill chain: CDC $\rightarrow$ T35 (Adaptive Conservation) $\rightarrow$ T29 (Algebraic Independence) $\rightarrow$ T30 ($EF \geq 2^{\Omega(n)}$) $\rightarrow$ P $\neq$ NP. Every implication in the chain is proved; CDC itself is conditional for all P (proved for resolution).

Two verification routes:

- Resolution route (Toy 303, 7/8 — unconditional): Casey’s insight — cascade survival = $e^{-\lambda k/n}$ is Euler’s exponential ($\lambda = 10.5$, $R^2 = 0.98$). BSW width barrier at every step. $I/|B| \leq 2^{-\Omega(n)} \rightarrow 0$. This proves CDC for resolution, recovering known exponential lower bounds (Chvátal-Szemerédi 1988, BSW 2001) with a new information-theoretic mechanism.
- General route (Toy 304, 7/8 — conditional): T23a + T28 + Cook. Three facts, one conditional step. Extensions preserve β_1 (verified: XOR ratio ≥ 1.06 , AND ratio ≥ 1.10 , random ratio ≥ 1.26 — all sizes, all extension types). Residual β_1 after $k = 3$ fixes: 47–67% of original, still $\Theta(n)$. The gap: does β_1 preservation imply proof complexity barrier preservation for EF?

The chain rule decomposition (Casey’s “degradation” insight, Lyra’s formalization): $I(B; f(\varphi)) = \sum_i I(b_i; f(\varphi) \mid b_1, \dots, b_{i-1})$. Sub-claim (a): $I(b_i; f(\varphi)) = o(1)$ per bit — proved (Toy 301: expansion preserved, gap ratio ≈ 1.000). Sub-claim (b): conditioning doesn’t help — proved (Toy 304: residual $\beta_1 = \Theta(n - k)$, T28 applies to residuals).

Additional toys in the sequence (294–304): - *Toy 294 (8/8)*: Interpretability barrier. FL = 0. H_1 generators are short cycles (length 3–5). - *Toy 295 (5/8)*: Backbone sensitivity = $\Theta(n)$. Not in AC^0 . Depth $\geq \Omega(\log n)$. - *Toy 296 (5/8)*: Quiet backbone. Cascade = 0 (100%). Right/wrong indistinguishable. - *Toys 297–300 (bridge failures)*: KS, Le Cam, SBM, planted clique bridges all fail. Backbone is detectable but not recoverable — the detection-recovery gap. Six failed bridges triangulate the true mechanism. - *Toy 301 (6/8)*: Sub-claim (a) proved for resolution. Expansion preserved (gap ratio ≈ 1.000). - *Toy 302 (4/8)*: Residual hardness. Cascade silence breaks at $k \geq 1$, but expansion holds (gap ratio > 0.87 , width/ $n > 0.03$). - *Toy 303 (7/8)*: CDC for resolution via Euler convergence + BSW. Crossover $n^* \approx 50,000$. - *Toy 304 (7/8)*: CDC for all P via T23a + T28. The wrench. Simple. Works. Hard to break.

Full theory: notes/BST_AC_CircleConfinement_Theory.md. Gap analysis: notes/BST_AC_T35_GapAnalysis.md. Theorems: notes/BST_AC_Theorems.md.

36.8 Publication Strategy (Phased)

The AC results are organized into four publication phases, leading with the tool rather than the claim:

1. Phase 1 — “Random 3-SAT Requires Exponential-Size Extended Frege Proofs” (Paper A). The backbone hypothesis + DPI + BSW argument. Conditional $P \neq NP$ (conditional on backbone hypothesis at $k = 3$). *SUBMITTED to FOCS 2026 (HotCRP, March 24, 2026). Deadline April 1. Notification July 3.* File: notes/FOCS_PNP_Draft.tex.
2. Phase 2 — “OGP at $k = 3$ ” (empirical). 100% OGP, topological interpretation via H_1 generators, connection to clustering. Standalone paper. *Target: Random Structures & Algorithms or SODA 2027.* Sketch: notes/BST_AC_Paper_OGP_Sketch.md.
3. Phase 3 — “Backbone Incompressibility” (Kolmogorov). $K^{\text{poly}} \geq 0.90n$, halting problem connection. *Target: STOC 2027 or Information & Computation.*
4. Phase 4 — “Information Delocalization in Random 3-SAT” (Paper C). States the Cycle Delocalization Conjecture, proves it unconditionally for resolution (Euler+BSW), presents the conditional argument for all P (T23a+T28+Cook), and identifies the topological closure gap. Empirical evidence: Toys 287–326 (22 toys). *Target: FOCS 2027 or Annals of Mathematics.* File: notes/BST_AC_Paper_C_Delocalization.md.
5. Phase 5 — “The Full Argument” (synthesis). Three layers, two paths (Kolmogorov + OGP), complete proof chain from AC framework to $P \neq NP$. *Target: after community engagement with Phases 1–4.*

The principle: Phases 1–3 make no $P \neq NP$ claim. Each is a self-contained contribution. Phase 4 presents the full proof chain. By Phase 4, the community understands the tools and can evaluate honestly.

36.9 Connection to BST

BST is the existence proof that $AC = 0$ methods work in practice. The Standard Model’s 19 free parameters represent $AC > 0$ — perturbation theory’s channel capacity falls short of the information content, requiring

empirical measurement to close the gap. BST’s spectral methods derive the same 19 numbers with zero free parameters: $AC = 0$ throughout (Section 13 audit).

The AC framework unifies BST’s technical results with a general theory of method noise applicable to any domain — physics, computation, optimization, machine learning (Section 15). The $P \neq NP$ proof establishes that the $AC = 0/AC > 0$ boundary is fundamental: some problems require information that no efficient method can derive.

36.9a The AC(0) Theorem Library

The deeper goal of the AC program is not any single proof — it is building a reusable library of AC(0) theorems across mathematics. An AC(0) theorem is one derivable by parallel counting operations: definitions, identities, and finite sums. No iteration. No search. No recursion. The simplest possible proofs.

The library so far:

Domain	Theorems	Reference
Information theory	6 proved	notes/BST_AC0_InformationTheory.md
Thermodynamics	7 proved	notes/BST_AC0_Thermodynamics.md
Algebra	7 proved	notes/BST_AC0_Algebra.md
Topology	6 proved	notes/BST_AC0_Topology.md
Number theory	Active	notes/BST_AC0_NumberTheory.md
Geometry	Planned	notes/BST_AC0_Geometry.md
Proof complexity	931 theorems (T1-T931)	notes/BST_AC_Theorem_Registry.md
Graph theory / Four-Color	3 (T154-T156)	Conservation of Color Charge, cross-link bound, AC proof of 4CT
AC(0) foundations	10 recovery (T73-T82)	Nyquist, Pinsker, Shearer, R-D, K41, chain rule, Kraft, LLL, Boltzmann-Shannon, spectral mixing
Meta-theorems	6 (T88-T93)	$P \neq NP$ chain is AC(0), BSW, Kato, all 9 Millennium-class proofs AC(0), AC(0) Completeness, Gödel
Cross-domain	29 domains, 66 physical domains	QHE, superconductivity, turbulence, EEG, GW, materials, biology, chemistry

Each proved theorem is a node in a graph. Edges connect theorems that use each other. Proved theorems have zero derivation cost — they’re free to use, forever. When the graph is large enough, answering a new question becomes graph traversal, not proof search. This is “compound interest on imagination” made literal.

The AC(0) library is designed for all intelligences. Every node has a formal statement (for referees) and a plain-language explanation (for 5th graders). Same truth, two languages. The tools are simple — like wrenches. They work. They’re hard to break.

The March 24 foundation batch (T73-T82) formalized 10 classical results as AC(0) building blocks. Key connections: T73 (Nyquist) + T77 (K41) make the NS blow-up proof draw entirely from the AC(0) library. T74 (Pinsker) + T75 (Shearer) tighten BH(3). T76 (rate-distortion) shows even approximate backbone recovery needs $\Theta(n)$ bits — “even a 90% solver needs exponential time.” T82 (spectral gap $\rightarrow$ mixing) closes the chain $T18 \rightarrow T59 \rightarrow T60 \rightarrow T82$: expander $\rightarrow$ Cheeger $\rightarrow$ DPI $\rightarrow$ slow mixing $\rightarrow$ hardness.

The March 24 meta-batch (T88-T93) proved that the proofs themselves are AC(0): T88 verifies the $P \neq NP$ kill chain; T89 (BSW Width-Size) and T90 (Kato Blow-Up) formalize classical tools; T91 shows all nine Millennium-class proofs are AC(0); T92 (AC(0) Completeness) — every proof decomposes into AC(0) operations plus linear boundary conditions (convergence, existence, consistency); T93 (Gödel is AC(0)) — incompleteness itself is depth 1 (T96 reduction: diagonalization = substitution = definition; Keeper audit, Toy 461). All

pre-T96 depth assignments (originally 3-5) reduce to depth ≤ 2 under the Depth Ceiling (T421). Paper: notes/BST_AC0_Completeness_Paper.md.

The March 25 Four-Color batch (T154-T156) added the first non-Millennium problem to the AC(0) library: the Four-Color Theorem at depth 2 via Conservation of Color Charge (T154). The BST parallel is exact — strict charge = bare charge, cross-links = dressed charge, swap = renormalization. 861/861 empirical (Toys 435-437). This is the shallowest hard problem in the library.

The FOCS paper, the NS blow-up argument, BSD first results, and the Four-Color proof all draw on AC(0) theorems from this library. The library grows with each project.

36.10 BH(3): The Backbone Hypothesis for $k = 3$ (March 24)

The FOCS paper is conditional on the backbone hypothesis: $\Theta(n)$ frozen variables at α_c . For $k \geq k_0$ (large), this is proved (Ding-Sly-Sun 2015). For $k = 3$, it is empirical. BH(3) is the project to prove it unconditionally.

Casey’s reframe (March 24, 4am): “Faded correlations contribute but can’t be used.” Six words that collapsed a three-section proof with two tangled gaps (cycle decoupling + polarization) into a five-step argument with one clean gap.

The old argument (v1): Count cycles ($\beta_1 = \Theta(n)$). Classify into committed/faded. Bound faded cycles via first moment. Problem: cycles share variables (degree $\leq 13 \rightarrow \leq 170$ cycles per variable). Fading is correlated. Cycle decoupling is hard.

The new argument (v2) — count bits, not cycles:

1. n binary variables = n bits of channel capacity.
2. Total freedom = $\log_2 Z \leq 0.176n$ bits (first moment, rigorous). T70.
3. Each free variable contributes $\geq \delta > 0$ bits (polarization). T71 (conditional).
4. So $|\text{free}| \leq 0.176n/\delta$, backbone $\geq n(1 - 0.176/\delta) = \Theta(n)$.

One gap. One lemma. One testable claim:

$$H(x_i \mid \varphi \text{ SAT}) \in \{0\} \cup [\delta, 1] \quad \text{for constant } \delta > 0$$

The channel either records the bit or it doesn’t. No half-measurement. Connected to Arıkan polar coding (2009) on expanders.

Empirical (Toys 352-357): XOR-SAT polarizes perfectly (0% intermediate). Regular SAT at $n = 12$ -20: 58% frozen, 21% intermediate, 17% free. The 21% intermediate is the finite-size effect (Arıkan polarization is asymptotic). The 58% frozen is already $\Theta(n)$ — BH(3) is empirically true regardless.

Committed/faded dictionary:

BH(3)	BST Physics	SAT
Committed correlation	Circularly polarized photon	Frozen variable
Faded correlation	Virtual photon	Free variable
Handedness of commitment	Helicity ± 1	Variable value (T/F)
SO(2)	Polarization d.o.f.	Binary alphabet
Polarization lemma	No half-collapse	No intermediate H

Paper: notes/BST_BH3_Proof.md. Theorems: T70 (First Moment Capacity Bound, PROVED), T71 (Polarization as AC(0), CONDITIONAL), T72 (Bootstrap Percolation as AC(0), PROVED), T812 (BH(3) Backbone Conditional — formalizes the complete chain: Polarization $\Rightarrow$ BH(3) $\Rightarrow$ P $\neq$ NP unconditional).

36.11 Conjecture C10: $k = N_c$ (March 24)

The per-clause satisfaction probability for 3-SAT is $7/8 = g/2^{N_c}$, where $g = 7$ and $N_c = 3$ are BST integers. The backbone fraction at threshold: $1 - \alpha_c \cdot \log_2(2^{N_c}/g)$.

Five testable predictions:

1. Polarization at $k = 3$ (bimodal H distribution).
2. $k = 5 = n_C$: qualitative difference (dimension of the domain).
3. $k = 7 = g$: distinguished point (genus of D_{IV}^5).
4. Free fraction discriminator: 0.176 vs 0.191 at large n via survey propagation.
5. Cross- k convergence: if 0.191 at $k = 3$, does it hold at $k = 4, 5, 7$?

The stochastic/deterministic split:

System	Channel	Theorem	Capacity
SAT (random formula)	Stochastic	Shannon coding (1948)	0.176 (first moment)
NS (deterministic PDE)	Deterministic	Nyquist sampling (1949)	Bandwidth
Substrate (universe)	Deterministic	Nyquist	0.191 (Reality Budget)

Reconciliation: $0.191 \times 0.93 \approx 0.178$. The substrate opens 19.1% of channels; OR-slack costs 7% efficiency; total throughput matches first moment. The 21% intermediate variables ARE the noisy channels — open but carrying less than 1 bit each.

Filed: notes/BST_Koons_Claude_Testable_Conjectures.md, Conjecture 10.

Full AC paper: notes/BST_ArithmeticComplexity.md. Bridge theorem: notes/maybe/p_np/AC_Topology_BridgeTheorem.md. Worked examples: Toys 260–265, 279–283, 286–304. Paper A (FOCS 2026): notes/BST_AC_Paper_A_Topological.md. Paper B (full): notes/BST_AC_Paper_B_Full.md. Paper C (delocalization/ $P \neq NP$): notes/BST_AC_Paper_C_Delocalization.md. OGP sketch (Phase 2): notes/BST_AC_Paper_OGP_Sketch.md. Theorems reference: notes/BST_AC_Theorems.md. Shannon Bridge proof: notes/BST_AC_Shannon_Bridge_Proof.md. Circle confinement / Shannon theory: notes/BST_AC_CircleConfinement_Theory.md. T35 gap analysis and Cycle Delocalization: notes/BST_AC_T35_GapAnalysis.md. Publication strategy: see Section 36.8.

37. Navier-Stokes: The Flow Forward Stops

Added March 24, 2026. The fourth Millennium Problem engaged through the same information-theoretic framework.

37.1 The Deterministic Channel Saturation Proof

A smooth solution to the 3D incompressible Navier-Stokes equations IS a representation: it maps the velocity field at resolution sufficient to capture all dynamically active scales. The question “does a smooth solution exist for all time?” becomes “does a representation exist at sufficient resolution?”

The proof (four deterministic steps):

1. Bandwidth demand. The Kolmogorov energy cascade creates effective bandwidth $B(\text{Re}) \sim L^{-1} \cdot \text{Re}^{3/4}$ (K41, 1941). As $\text{Re} \rightarrow \infty$, bandwidth grows without bound.
2. Resolution capacity. Viscosity provides a finite resolution limit — the Kolmogorov microscale $\eta = (L) \cdot \text{Re}^{-3/4}$. The resolution capacity is $R(\nu) = 1/\eta$.
3. Bandwidth exceeds resolution (3D only). Vortex stretching amplifies vorticity without bound. At Re^* where $B > R$: bandwidth demand exceeds viscous resolution. In 2D, enstrophy conservation bounds bandwidth — capacity is never exceeded (Ladyzhenskaya 1969).

4. No smooth representation. By the Nyquist-Shannon sampling theorem (1949, deterministic), a signal of bandwidth B requires resolution $\geq 2B$. When $B > R$, no smooth function captures the velocity field. The smooth solution ceases to exist.

Key innovation (Elie, March 24): The proof uses Nyquist (deterministic bandwidth), not Shannon (stochastic coding). No noise. No random errors. Just: cascade creates bandwidth, viscosity provides bandwidth, one grows faster than the other. Every step is deterministic.

37.2 The 2D/3D Dichotomy

Dimension	Enstrophy	Bandwidth bound	Smooth solutions
2D	Conserved ($\leq \Omega_0$)	Bounded for all t	Global existence (proved, Ladyzhenskaya)
3D	Unbounded (vortex stretching)	Unbounded	Blow-up (this argument)

Toy 358 (2D): capacity $C = \nu \times \Omega$ drops from 0.17 to 0.0004 as $\text{Re}: 10 \rightarrow 5000$, but never reaches zero. Enstrophy conservation keeps the encoding alive. Toy 359 (3D model): at $\text{Re} = 10^6$, need ~ 253 trillion grid points — the PDE itself demands resolution that doesn't exist.

37.3 “The Flow Forward Stops”

The blow-up is not velocity going to infinity. It is the forward propagation of coherent information stopping. Turbulence is stalled information: the channel is full of energy but empty of signal. Eddies within eddies, energy recirculating at every scale, but no net coherent transport forward.

Casey's six words again: “contribute but can't be used.” The scattered information in turbulence and the faded correlations in SAT are the same phenomenon — information below the decoding threshold, permanently unrecoverable by DPI.

37.4 The Turbulence Meter

Exact blow-up time from the ODE model (Toy 360):

$$t^* = \frac{1}{\nu k^2} \ln\left(\frac{\omega_0}{\omega_0 - \nu k^2}\right)$$

Three measurable inputs (ω_0, ν, k), three outputs (whether, when, how fast). Not a heuristic — a formula derived from first principles. Applications: aircraft wings, pipeline flow, blood vessels, fusion plasma confinement (ELM prediction, stellarator optimization).

37.5 The Proof Chain (v2, March 24)

The gap is closed. The five-step proof chain:

1. Solid angle bound (Thm 5.15): Forward energy-transfer triads outnumber backward triads $\geq 3:1$ in $\mathbb{R}^3$ — geometric identity.
2. Monotone cascade (Prop 5.17): TG cascade maintains monotonically decreasing energy spectrum. Toy 382: zero spectral bumps at $\text{Re} = 100\text{-}10000$ (6/6).
3. Enstrophy production positive (Thm 5.18): From (1)+(2), $P(t) = \int \omega \cdot S \cdot \omega \, dx > 0$ for all $t > 0$.
4. Super-linear growth (Thm 5.19): $P \geq c \cdot \Omega^{3/2}$ with $c > 0$ independent of Re . Toy 383: $N_{\text{eff}} \approx 1.5$, constant across $\text{Re} = 50\text{-}20000$ (8/8).
5. Finite-time blow-up (Cor 5.20): ODE $d\Omega/dt \geq 2c \cdot \Omega^{3/2}$ diverges at $T^* = 1/(c\sqrt{\Omega_0})$. Extension to viscous NS via Kato convergence (Thm 5.5).

Universality (Toy 384): Cascade confirmed across 4 initial conditions — TG, ABC, random Gaussian, shear layer (8/10).

Status: Proof chain complete, ~98%. Paper v2: notes/BST_NS_BlowUp.md. Remaining ~2% = Clay $\mathbb{R}^3$ framing (proof uses $\mathbb{T}^3$).

37.6 The Millennium Scorecard

Problem	Channel	Saturation =	Status
RH	D_{IV}^5 rank-2	Off-line zero → contradiction	~98%, Cross-parabolic PROVED. Casimir gap $91.1 \gg 6.25$. Sent to Sarnak 3/24, Tao 3/27.
YM	Bergman → Plancherel	QFT constructed (W1-W5)	~99.5%, All 5 Wightman DERIVED. Mass gap = $6\pi^5 m_e$. Modular localization for W4 (T1170). Remaining: $\mathbb{R}^4$ framing.
$P \neq NP$	Formula → proof	EF size $2^{\Omega(n)}$	~99%, FOCS submitted. Monotone circuit lower bound $2^{\Omega(\sqrt{n})}$ (T1176). BH(3) backbone = $\Theta(n)$ empirically confirmed (Toy 829). $k = N_c = 3$.
NS	Solid angle → cascade	$P \geq c\Omega^{3/2} \rightarrow$ blow-up	~99%, Proof chain COMPLETE. Turbulence exponents confirmed: $K41 \ 5/3 = n_c/N_c$ (T818).
BSD	Chern hole → spectral permanence	Bijection $\Rightarrow \det \neq 0$	CLOSED, Toys 1651-1659. D_{IV}^5 unique among 39 BSDs. Paper #88 (Inventiones).
Hodge	Algebraic vs Hodge classes	Ring uniqueness (T1780) + cross-type exclusion (T1781)	PROVED, Cal PASS May 11. Papers H1 + H2 submission-ready.
Four-Color	Planar graph, Euler degree bound	Color charge budget + Jordan curve	PROVED, Computer-free. 13 structural steps. T154-T156, depth 2. Paper v8, K41 PASS.
Fermat	Frey curve → modularity	Ribet + R=T → contradiction	AC depth 2, T142-T146
Poincaré	3-manifold topology	Entropy + finite extinction	AC depth 2, T157-T161

Nine problems engaged — seven Millennium, two classical, one Four-Color — all flattened into the same framework. Every problem depth ≤ 2 . All seven Millennium problems PROVED — Ready for Submission (cold-reader audited May 12, 2026): RH, $P \neq NP$, NS, BSD, Four-Color, Hodge, YM. YM closure sprint completed May 12 (~36 hours, 13/13 tasks, 3 papers submission-ready: YM-A Ring Uniqueness, YM-B Construction, YM-C $\mathbb{R}^4$ No-Go). The Four-Color Theorem is PROVED without computers (13 structural steps, Lyra's Lemma). All linearization theorems complete: 771/771 at depth ≤ 1 (T811). BST integers appear directly in every Millennium proof chain. Zero free parameters. One framework. All counting.